

Segmentation Engine Documentation

1 Overview

The segmentation engine is a C++ HTTP service that performs route analysis and segment extraction. The server loads its configuration from `config/settings.json` and exposes a set of REST endpoints.

2 Building and Running

```
cmake -S . -B build
cmake --build build
./build/segmentation_engine % binary name may vary
```

The default server port and endpoint list are defined in `config/settings.json`.

3 API

3.1 POST /upload

Save a client-provided map JSON file.

Body JSON object representing the map data.

Response { "ok": true, "file": "uploads/map_<timestamp>_<rand>.json" }

3.2 GET /view?map=<path>

Render an interactive HTML viewer for a previously uploaded map.

Query map – path returned from /upload (must be under uploads/).

Response HTML page comparing raw OSRM geometry to derived route signal.

3.3 GET /viewLab?map=<path>

Redirect to the static Signal Lab UI.

Query optional map path.

Response HTTP 302 redirect to `/static/signal_lab.html`.

3.4 GET /lab/meta?map=<path>

List available uploads and optionally preload one map.

Query optional map path.

Response JSON { "uploads": ["uploads/..."], "preload": "uploads/..."? }.

3.5 POST /lab/resample

Build a route signal and resample requested variables onto a uniform distance grid.

Body { "map": "uploads/...", "ds_m": <step_m>, "vars": ["elev", ...] }

Response { "ok": true, "s_km": [...], "ds_m": <step_m>, "series": "var": [...]}

3.6 POST /wavelet

Run wavelet-based terrain segmentation on a map and optionally persist results.

Body { "map": "uploads/...", "fn": "terrain", "persist": false }

Response JSON containing **segments** with segment geometry and metadata, **series** with processed signals, **s_km_uniform** axis and **ds_m** spacing. If database persistence fails, a **db_error** field is included.

3.7 GET /segments

Fetch previously persisted segments intersecting a bounding box.

Query either **bbox=west,south,east,north** or **minLon, minLat, maxLon, maxLat**.

Response GeoJSON FeatureCollection of matching segments.